

Amivantamab Plus Lazertinib vs Osimertinib in First-line *EGFR*-mutant Advanced NSCLC

Final Overall Survival from the Phase 3 MARIPOSA Study

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DECLARATION OF INTERESTS

James Chih-Hsin Yang

Consulting or Advisory Role: Boehringer Ingelheim, Novartis, AstraZeneca, Clovis Oncology, MSD Oncology, Celgene, Bayer, Pfizer, Ono Pharmaceutical, Bristol Myers Squibb, Yuhan, Hansoh, Blueprint Medicines, Daiichi Sankyo, G1 Therapeutics, AbbVie, Takeda, Amgen, Incyte, Eli Lilly, GSK, Merck KGaA, Daiichi Sankyo/AstraZeneca, Puma Biotechnology, Gilead Sciences, Taiho Pharmaceutical, Bayer, Roche/Genentech, Sanofi, ArriVent Biopharma

Honoraria: Boehringer Ingelheim, Roche, Merck Sharp & Dohme, AstraZeneca, Novartis, Bristol Myers Squibb, Ono Pharmaceutical, Takeda, Eli Lilly, Pfizer, Amgen, AstraZeneca/MedImmune, Dizal Pharma, Taiho Pharmaceutical, Roche/Genentech, Daiichi Sankyo/AstraZeneca, MSD Oncology, BeiGene, Gilead Sciences, Sanofi/Regeneron

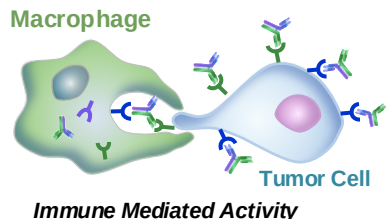
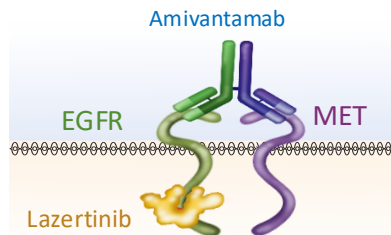
Travel, accommodations, expenses: Pfizer, AstraZeneca, Dizal Pharma

Research funding: AstraZeneca

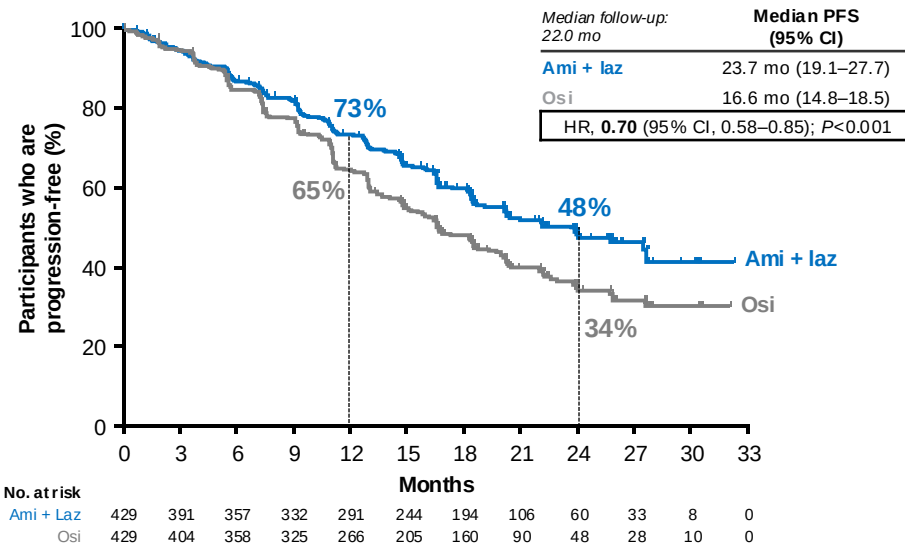


Background

- In MARIPOSA, 1L amivantamab + lazertinib significantly improved PFS vs osimertinib^{1,2}
- Amivantamab + lazertinib is approved for patients with 1L EGFR-mutant advanced NSCLC^{3,4}
- 1L amivantamab + lazertinib exhibits a triple mechanism of action with a reduction in the spectrum and complexity of acquired resistance⁵



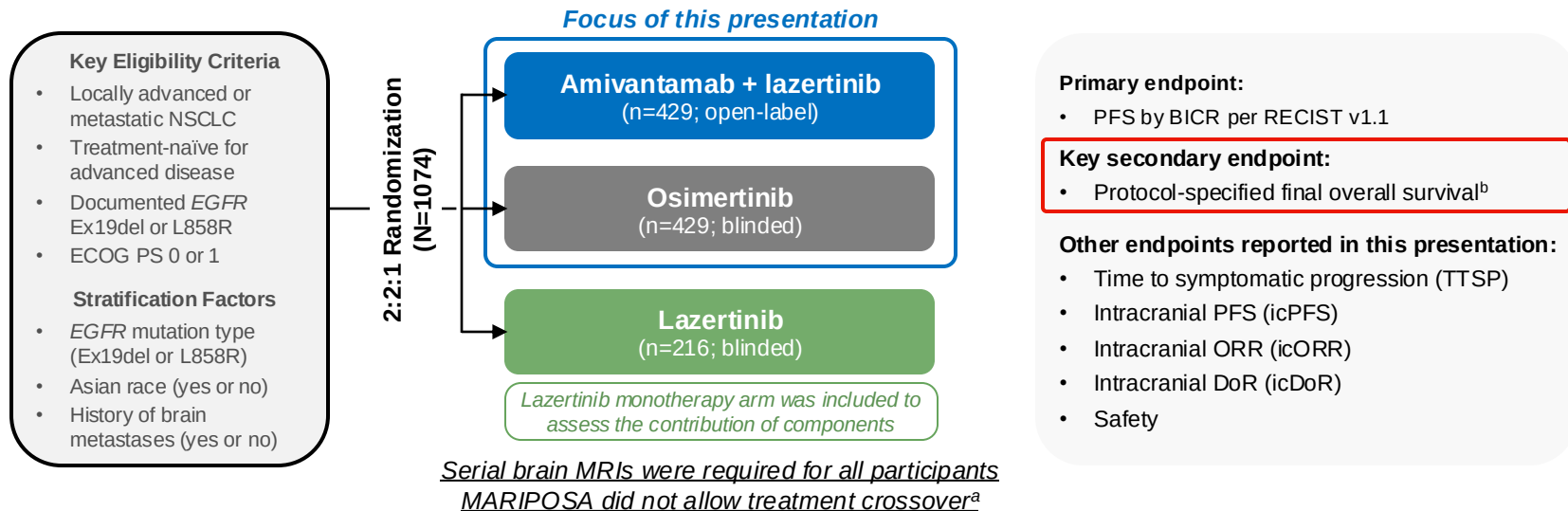
1L Amivantamab + Lazertinib Primary Endpoint: PFS by BICR^{1,2}



Here, we report the protocol-specified final overall survival results of
1L amivantamab + lazertinib vs osimertinib from MARIPOSA



Phase 3 MARIPOSA Study Design

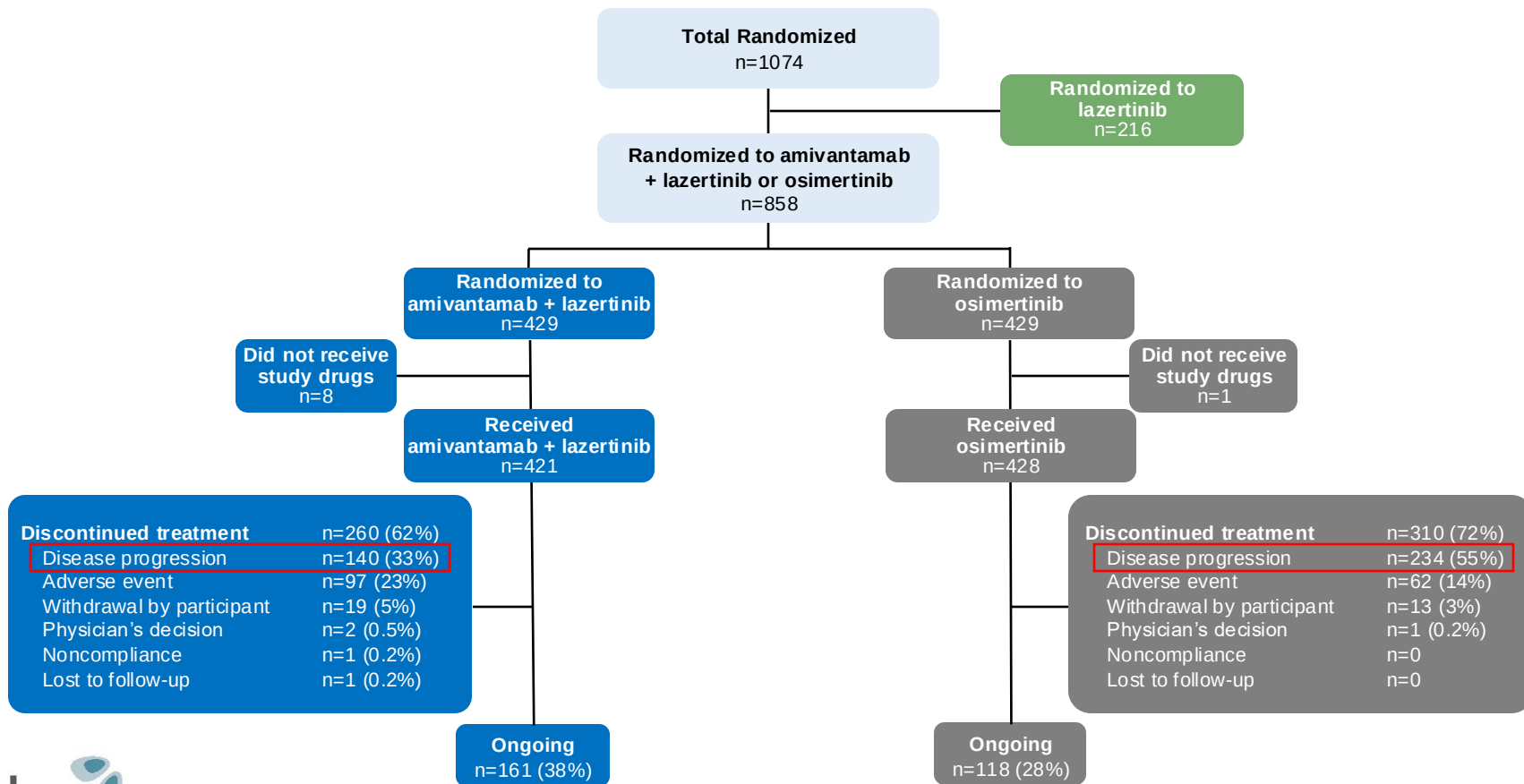


OS was a key secondary endpoint with prespecified alpha to assess significance

- Protocol-specified final OS analysis was planned for when ~390 deaths had occurred in the amivantamab + lazertinib and osimertinib arms
- OS was tested with a 2-sided alpha of 0.05, determined by O'Brien-Fleming alpha spending approach as implemented by the Lan-DeMets method
 - In the prespecified interim analysis, a 2-sided alpha of 0.005 was allocated for OS
 - The protocol-specified final analysis of overall survival was subsequently evaluated at a 2-sided significance level of 0.0484



Participant Disposition



Baseline Demographics and Clinical Characteristics^{1,2}

Baseline characteristics were well balanced across both arms

Characteristic, n (%)	Amivantamab + lazertinib (n=429)	Osimertinib (n=429)
Median age, years (range)	64 (25–88)	63 (28–88)
Female	275 (64)	251 (59)
Race		
Asian	250 (58)	251 (59)
White	164 (38)	165 (38)
Other ^a	15 (3)	13 (3)
ECOG PS 1	288 (67)	280 (65)
History of smoking	130 (30)	134 (31)
History of brain metastases	178 (41)	172 (40)
EGFR mutation type ^b		
Ex19del	258 (60)	257 (60)
L858R	172 (40)	172 (40)
Adenocarcinoma subtype	417 (97)	415 (97)

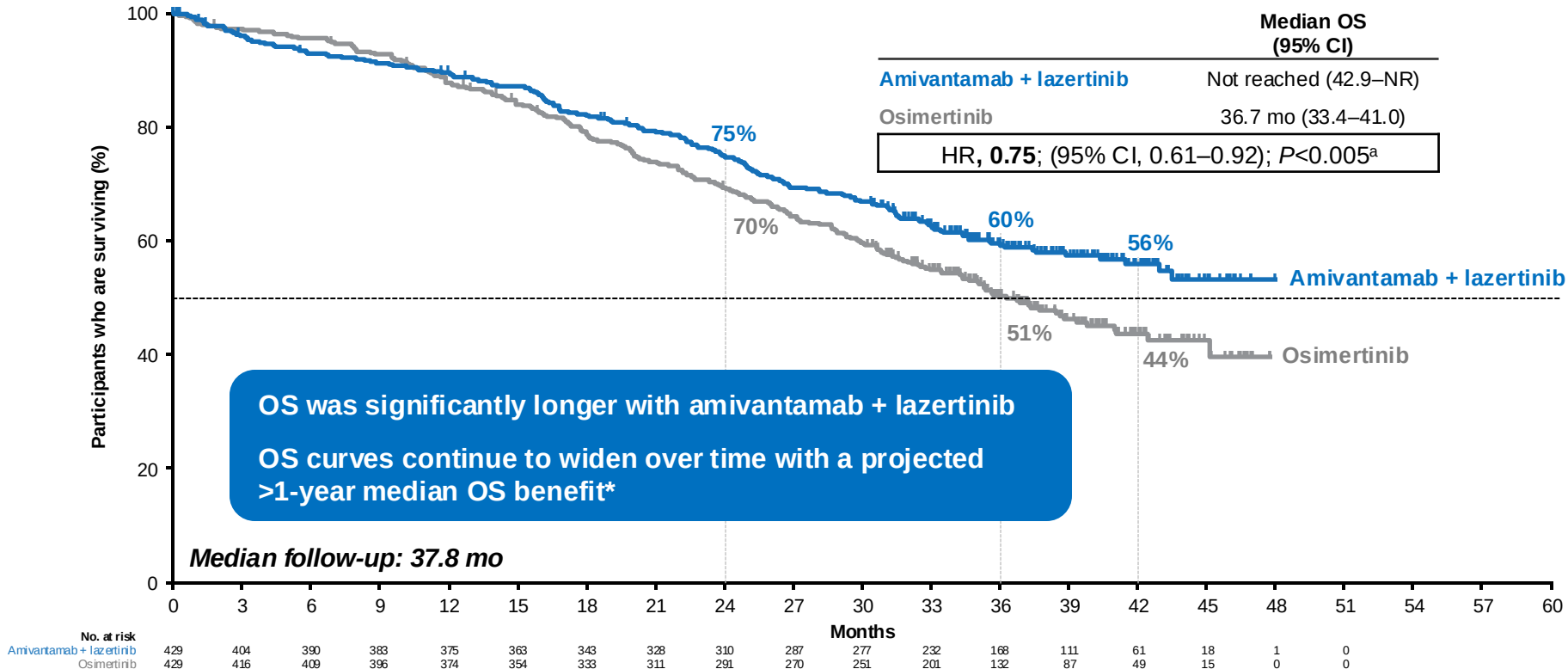
Note: percentages may not sum to 100 due to rounding.

^aOther includes American Indian or Alaska Native, Black or African-American, multiple, and unknown. ^bOne patient in the amivantamab + lazertinib arm had both Ex19del and L858R.

1. From *New England Journal of Medicine*, Cho BC, et al., Amivantamab plus Lazertinib in Previously Untreated EGFR-Mutated Advanced NSCLC, Volume No. 391 (16), Pages 1486-1498. Copyright © (2024) Massachusetts Medical Society. Reprinted with permission from Massachusetts Medical Society. 2. Cho BC, et al. Presented at the European Society for Medical Oncology (ESMO) Congress; October 20-24, 2023; Madrid, Spain.



MARIPOSA: Overall Survival



***Based on an exponential distribution assumption of OS in both arms, the improvement in median OS is projected to exceed 1 year.**

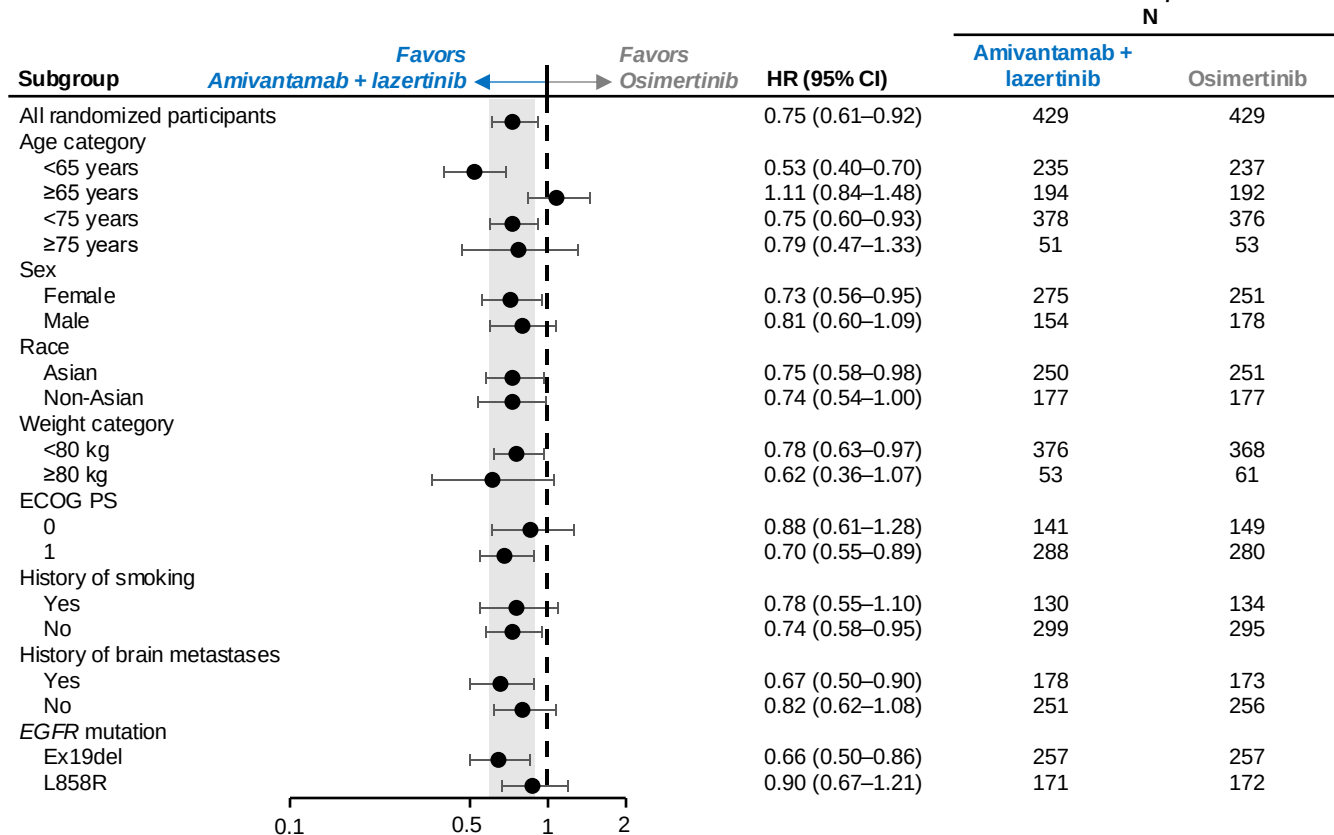
Note: Last participant was enrolled in May 2022. Clinical cutoff date was December 4, 2024. In total, 390 deaths had occurred in the amivantamab + lazertinib (173 deaths) and osimertinib (217 deaths) arms.

***P-value** was calculated from a log-rank test stratified by mutation type (Ex19del or L858R), race (Asian or Non-Asian), and history of brain metastasis (present or absent). Hazard ratio was calculated from a stratified Cox regression model.



Overall Survival in Predefined Subgroups^a

A generally consistent OS benefit for amivantamab + lazertinib over osimertinib was observed across predefined subgroups

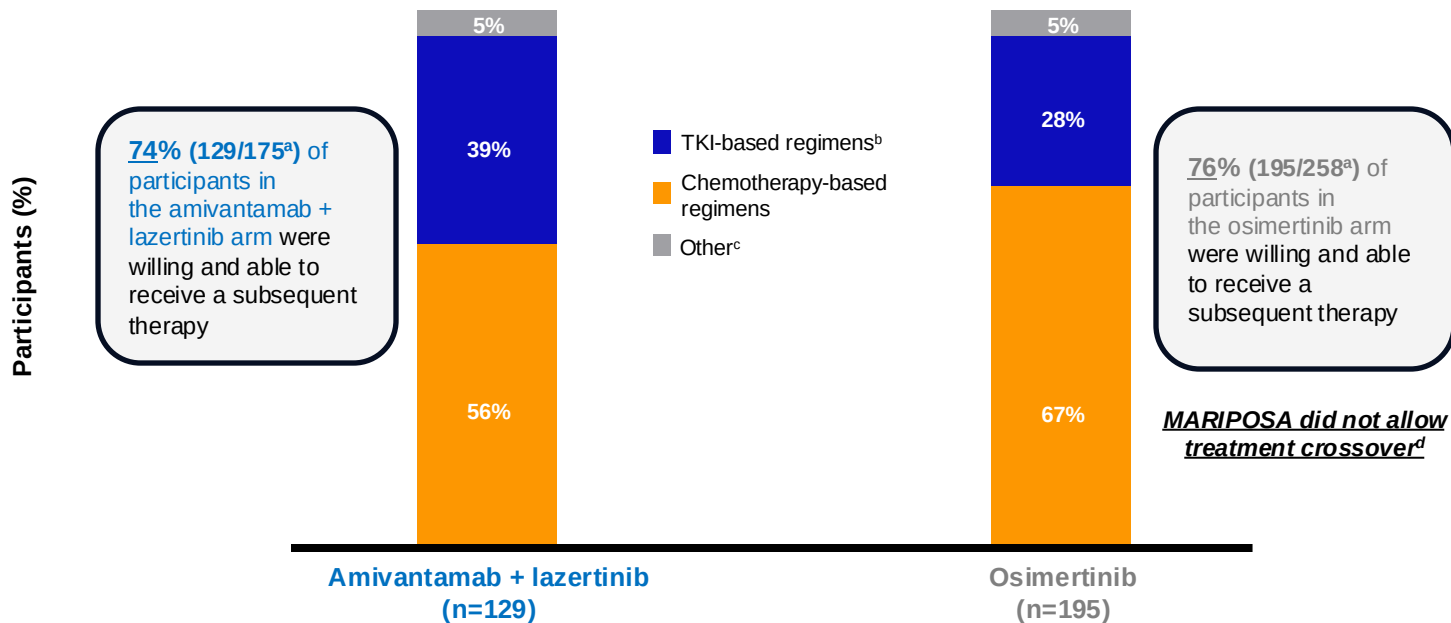


Note: Gray box indicates 95% CI of HR for all randomized participants. ^aSubgroup analyses were not part of the hypothesis testing of the trial and should not be used to infer definitive treatment effects.



First Subsequent Therapy

Most common subsequent therapy class was chemotherapy-based regimens in both arms



74% received 2L therapy, suggesting a long-term treatment plan after 1L amivantamab + lazertinib is feasible

Percentages may not total 100 due to rounding.

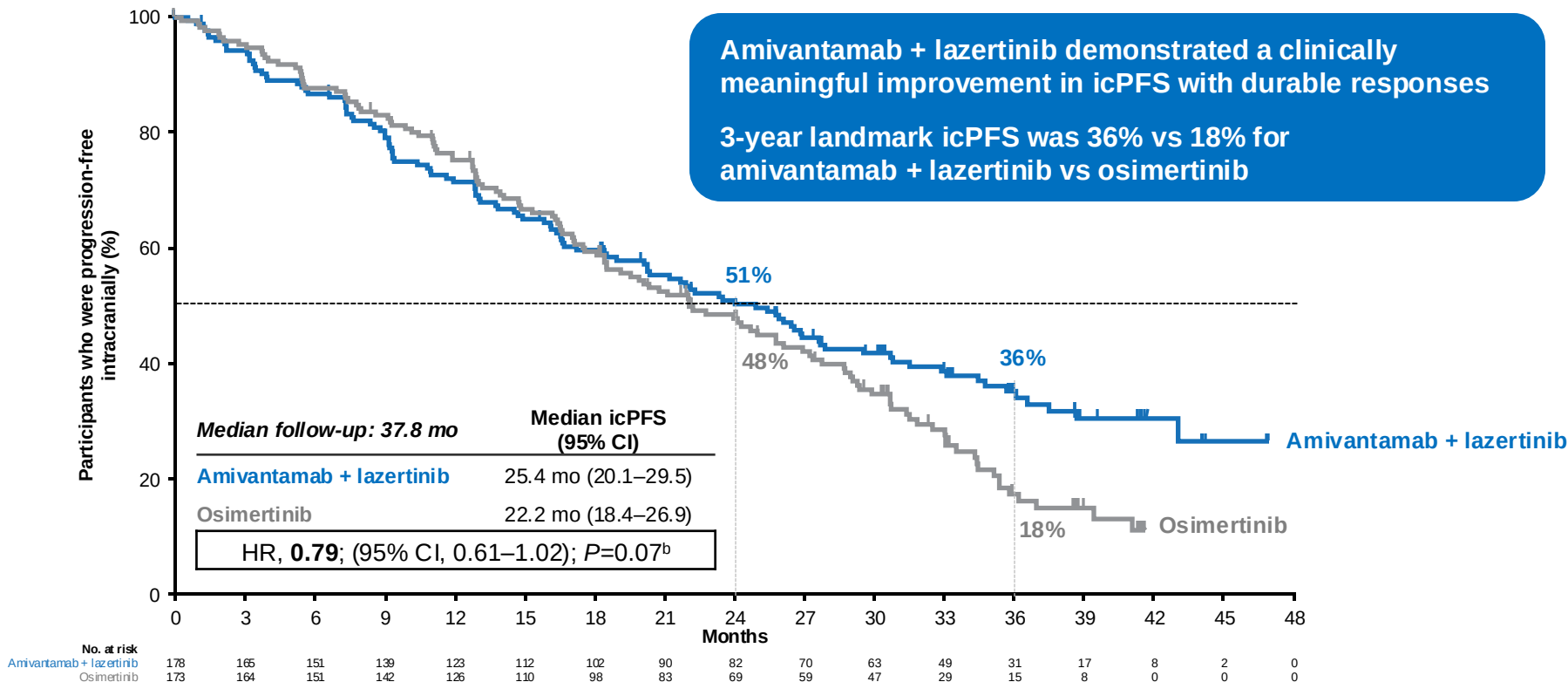
^aDenominator is the number of participants who had disease progression and discontinued randomized treatment. ^bTKI-based regimens include TKI + chemotherapy (5% in both arms). ^cOther therapy included VEGFi alone, IO alone, herbals, antibody-drug conjugates, ALK tyrosine kinase inhibitors, c-MET tyrosine kinase inhibitors, amivantamab (1 participant received amivantamab-chemotherapy after amivantamab-lazertinib; after osimertinib, 1 participant received amivantamab-chemotherapy, 1 participant received amivantamab-lazertinib, and 1 participant received amivantamab monotherapy), and investigational agents. ^dMARIPOSA did not allow crossover as amivantamab-based regimens were not approved in the 2L setting during MARIPOSA enrollment.



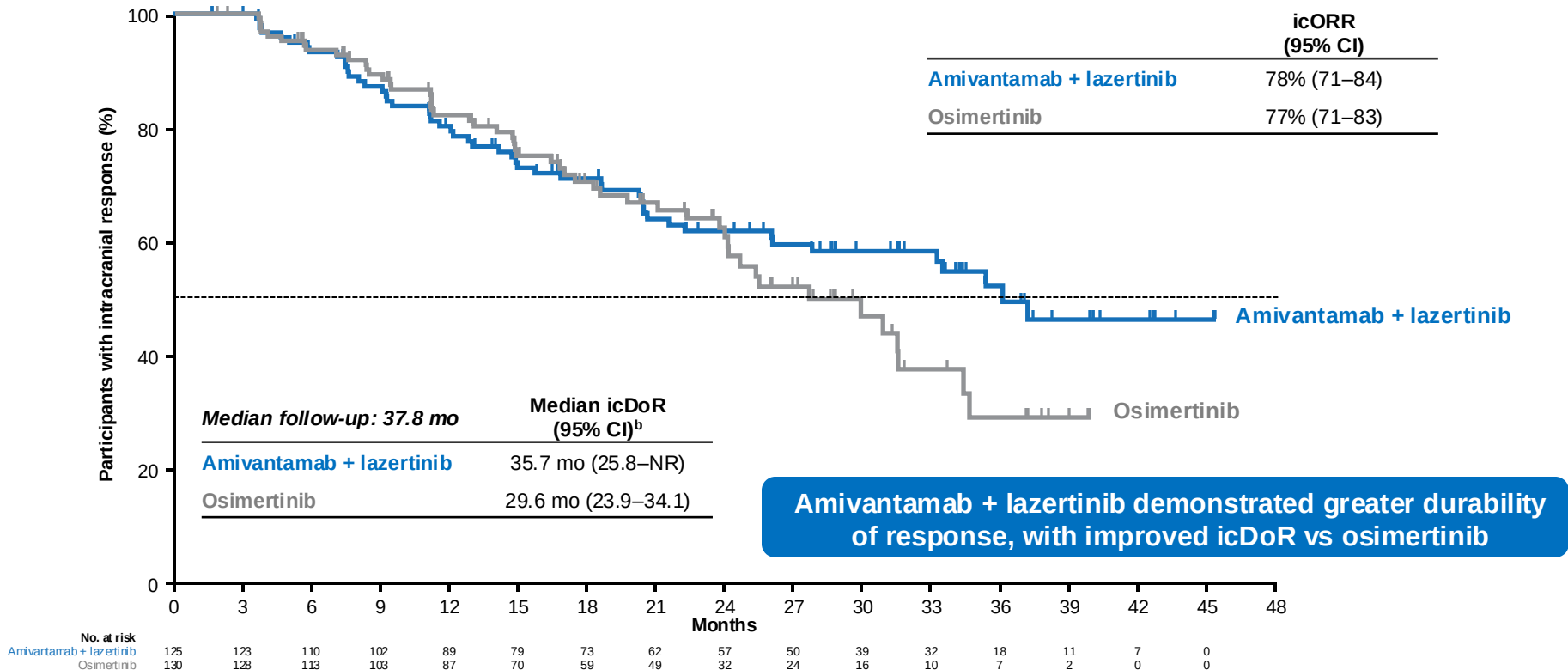
Intracranial PFS^a

Amivantamab + lazertinib demonstrated a clinically meaningful improvement in icPFS with durable responses

3-year landmark icPFS was 36% vs 18% for amivantamab + lazertinib vs osimertinib

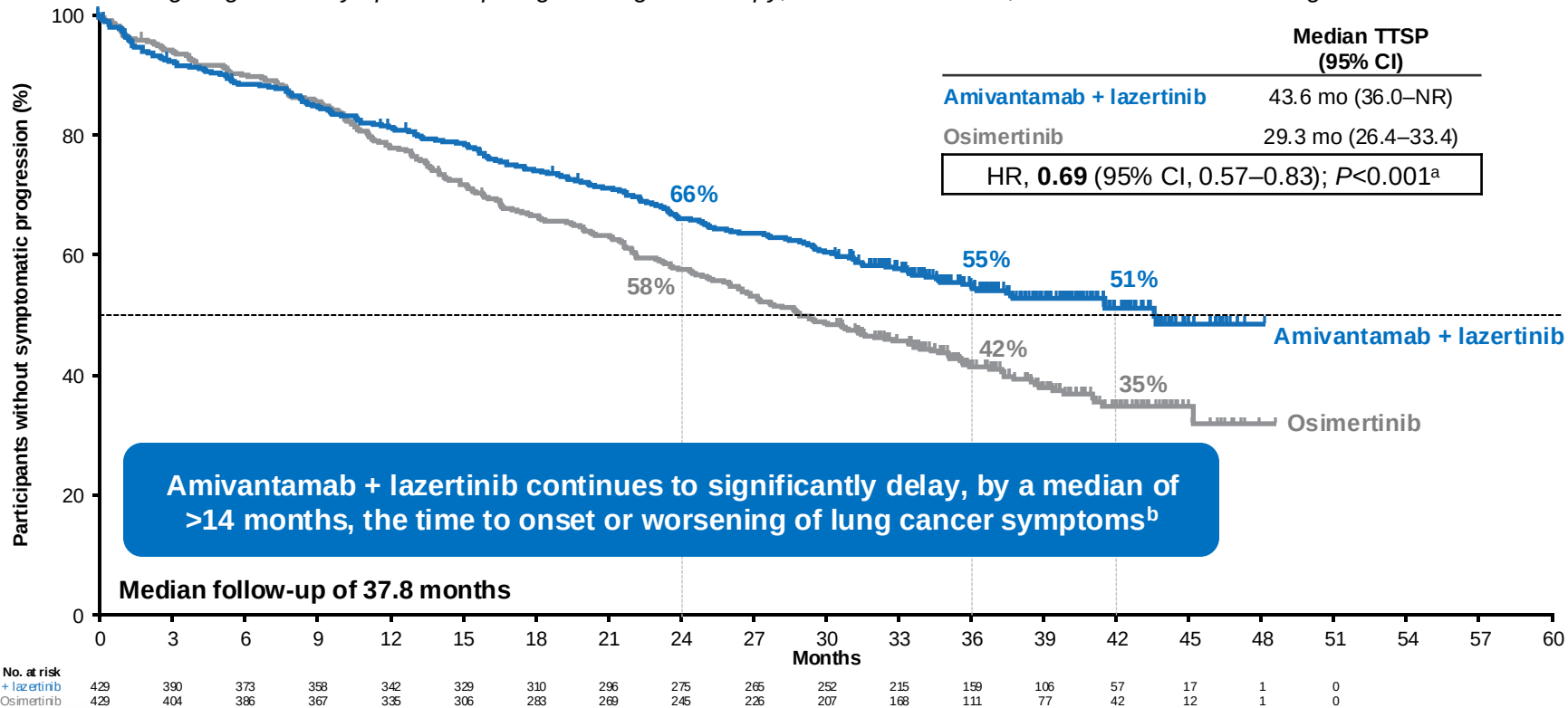


Intracranial DoR^a



Time to Symptomatic Progression (TTSP)

Symptomatic progression is a patient-relevant endpoint that measures time from randomization to the onset of new/worsening lung cancer symptoms requiring a change in therapy, clinical intervention, or death based on investigator discretion



Safety

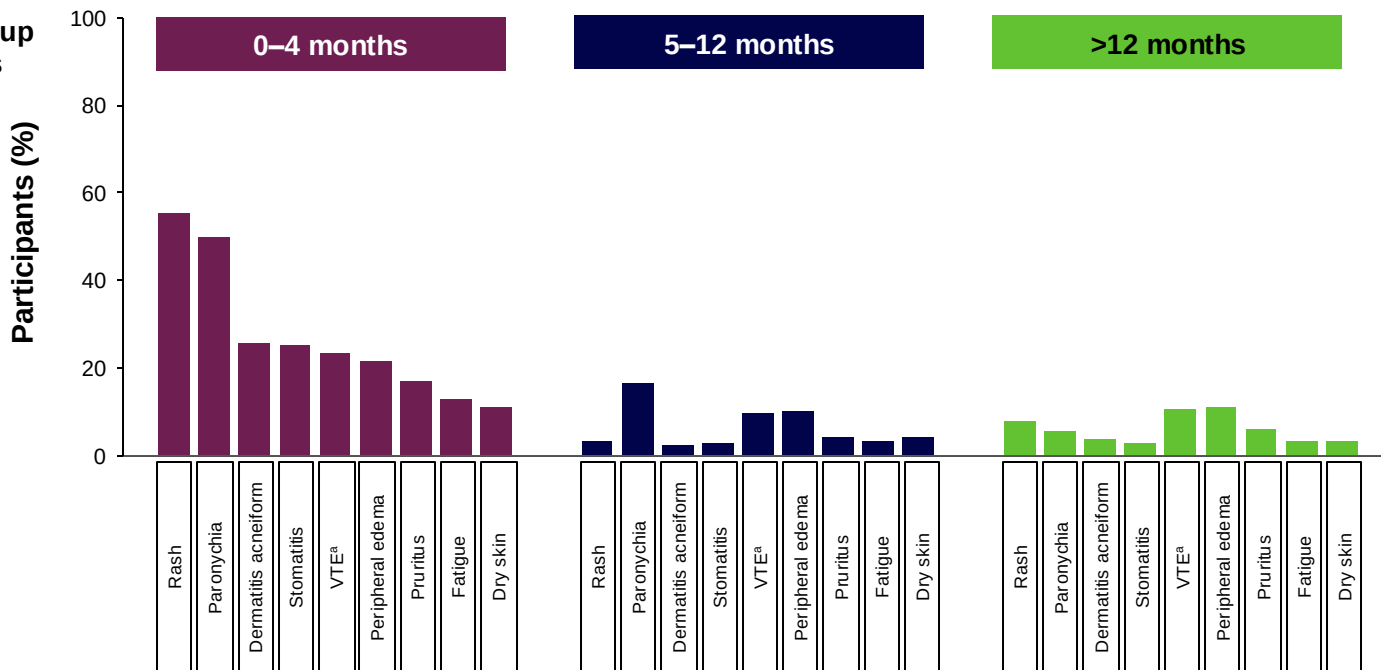
- Median duration of treatment was **27.0 mo** for amivantamab + lazertinib and **22.4 mo** for osimertinib
- Safety profile was consistent with the primary analysis¹
 - AEs were mostly EGFR- and MET-related and grades 1–2^{1,2}
- A minority of participants were prescribed antibiotics for rash (21%) at study initiation²
- Few were on anticoagulation (5%) at baseline², with VTE^a occurring in **40%** in the amivantamab + lazertinib arm and **11%** in the osimertinib arm

AEs by preferred term (≥20% of participants in either group)	Amivantamab + lazertinib (n=421)		Osimertinib (n=428)	
	Any grade	Grade ≥3	Any grade	Grade ≥3
Related to EGFR inhibition				
Paronychia	291 (69)	49 (12)	127 (30)	2 (<1)
Rash	271 (64)	73 (17)	136 (32)	3 (<1)
Diarrhea	133 (32)	9 (2)	200 (47)	4 (<1)
Dermatitis acneiform	127 (30)	37 (9)	55 (13)	0
Stomatitis	126 (30)	5 (1)	92 (21)	1 (<1)
Pruritus	107 (25)	2 (<1)	75 (18)	1 (<1)
Related to MET inhibition				
Hypoalbuminemia	216 (51)	26 (6)	29 (7)	0
Peripheral edema	162 (38)	8 (2)	29 (7)	1 (<1)
Other				
Infusion-related reaction	275 (65)	27 (6)	0	0
ALT increased	170 (40)	28 (7)	66 (15)	8 (2)
AST increased	139 (33)	15 (4)	68 (16)	6 (1)
Constipation	130 (31)	0	70 (16)	0
COVID-19	125 (30)	8 (2)	112 (26)	9 (2)
Anemia	114 (27)	20 (5)	112 (26)	10 (2)
Decreased appetite	114 (27)	4 (1)	84 (20)	7 (2)
Nausea	99 (24)	5 (1)	65 (15)	1 (<1)
Hypocalcemia	96 (23)	11 (3)	37 (9)	0
Asthenia	84 (20)	13 (3)	54 (13)	7 (2)
Muscle spasms	84 (20)	3 (<1)	36 (8)	0
Thrombocytopenia	74 (18)	4 (1)	92 (21)	6 (1)



First Onset of Key AEs for 1L Amivantamab + Lazertinib

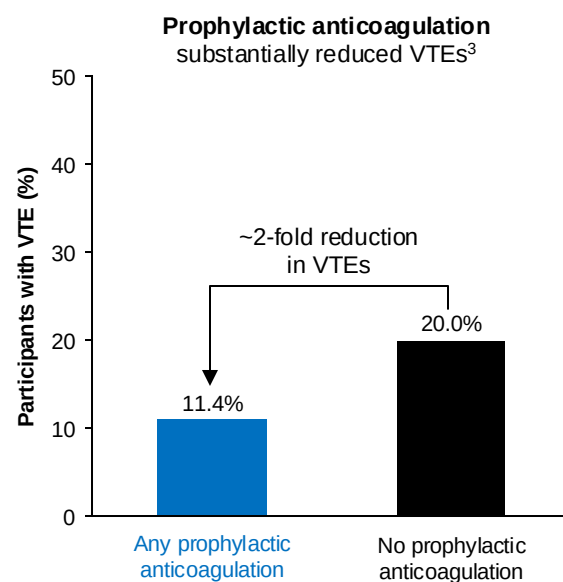
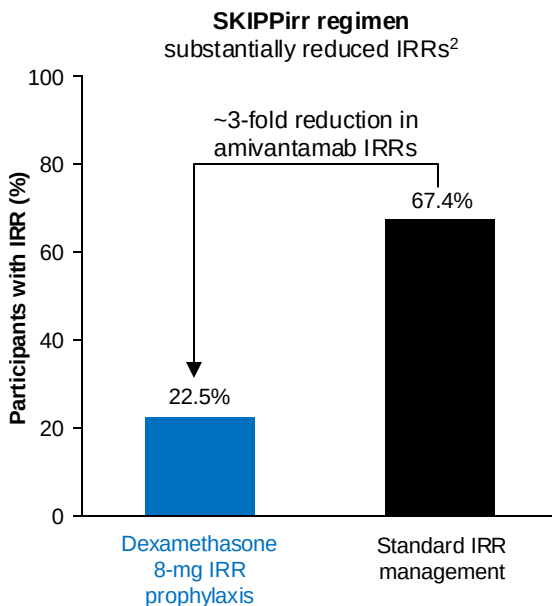
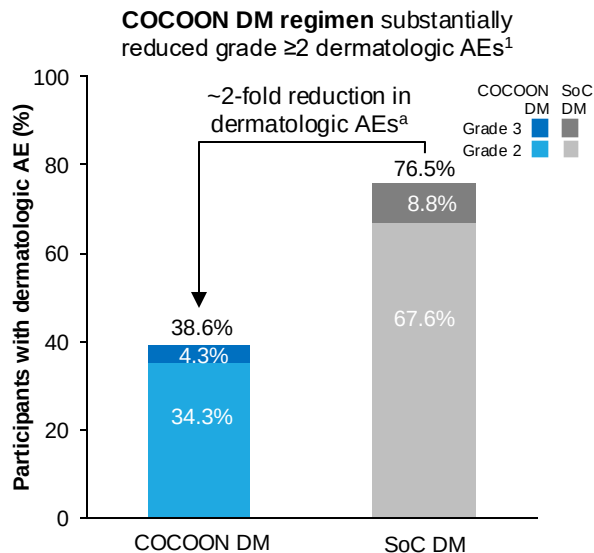
Median follow-up
of 37.8 months



Most first onset AEs occur early (0-4 months), with longer-term follow-up showing no new safety signals and indicating that long-term treatment is feasible



Early Onset AEs can be Significantly Reduced with Prophylactic Approaches



Data from COCOON will be presented on
Thursday, 27 March 2025 at 16:50–16:55 CET

Early onset AEs can be reduced using simple and accessible preventative approaches



Conclusions

- 1L amivantamab + lazertinib led to a statistically significant and clinically meaningful reduction in mortality vs osimertinib (HR, **0.75**; $P < 0.005$) in participants with previously untreated *EGFR*-mutant advanced NSCLC
 - A **>12-month median OS benefit** is projected for amivantamab + lazertinib versus osimertinib^a
 - **60% of participants were alive at 3 years in the amivantamab + lazertinib arm** vs 51% for osimertinib; benefit continued at 42-months with survival rates of 56% and 44%, respectively
- **Twice as many** participants receiving amivantamab + lazertinib were **intracranially progression-free** at 3 years (36% vs 18%) with a **longer intracranial DoR** vs osimertinib (35.7 vs 29.6 months)^b
- Amivantamab + lazertinib delayed the time to a participant experiencing symptoms from their lung cancer by a median of >14 months (TTSP; $P < 0.001$)
- AEs with 1L amivantamab + lazertinib occurred early; prophylactic interventions have now been shown to substantially **reduce the incidence of these key AEs** (dermatologic AEs, IRRs, and VTE)



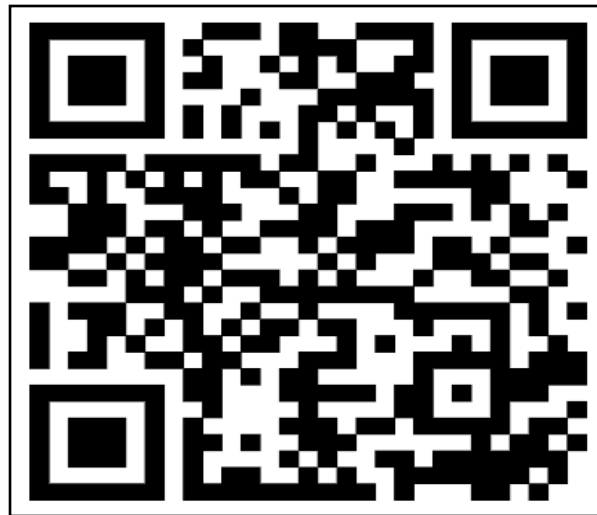
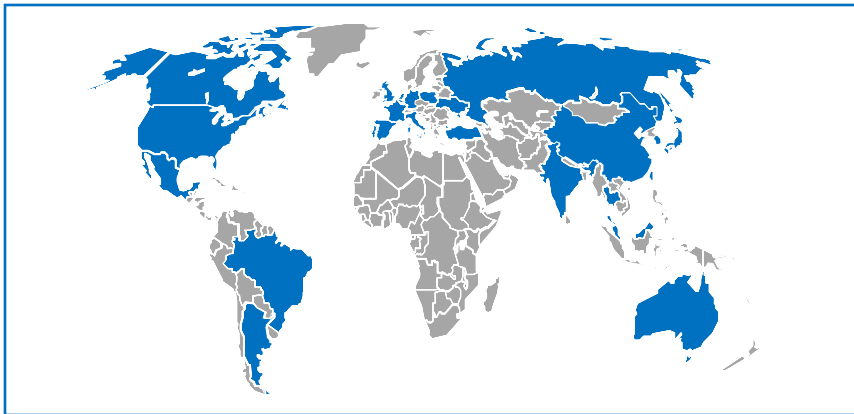
Patients live longer with 1L amivantamab + lazertinib, with MARIPOSA demonstrating practice-changing superior OS versus osimertinib and extending median survival beyond 3 years



Acknowledgements

- Participants who were enrolled in the study and their families and caregivers
- Physicians and nurses who cared for participants and staff members who supported this clinical trial
- Staff members at the study sites and involved in data collection/analyses
- Medical writing assistance was provided by Humanity Communications Inc. and funded by Johnson & Johnson

A total of 1074 participants from 28 countries
randomized in the MARIPOSA study



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